This Course

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UAB-IDEA

September 2026

Section 1

This Course

Schedule

1 to 4 September 10:00 to 12:30

7 to 9 September 10:00 to 12:30

Final exam. Thursday 10 September. It covers the core material up to and including 9 September; marked optional material is excluded unless explicitly announced.

Objectives

By the end of the brush-up, you should be better able to identify where you need to consolidate your understanding, fill gaps, or learn more, so that you can engage more effectively with the formal first-year courses that follow.

- Use notation and definitions to locate concepts that need review.
- Work through representative problems to see which techniques are secure and which need more practice.
- Check dimensions, assumptions, and proof steps to identify gaps in reasoning.
- Leave with a concrete list of topics to revisit before and during the first-year courses.

Syllabus

01 Preliminaries

02 Linear Algebra

03 Functions

04 Differentiation

05 Convexity & Concavity

06 Optimization

07 Integrals

08 Statistics

How the materials are organised

Core material

Unboxed material contains the main review and worked examples. We prioritize it in class.

Blue boxes

Optional proofs, and technical details. These may be skipped without losing the tools required for the course.

Red boxes

Additional examples and optional applications. These may be skipped or used for extra practice.

Exam scope. The exam is based on the core material.

Blue-box and red-box material is optional and will not be examined in this brush-up course. Most of it will be covered in later IDEA courses, so it is still worth previewing if you have time.

Evaluation

Not on IDEA report The grade does not appear on your IDEA report and does not count toward your GPA.

Still graded Your work will still be graded.

For self-assessment Use your results to identify areas that may need further revision.

Resources

LECTURE NOTES & TUTORIALS

- ♦ **M. J. Osborne**: *Mathematical Methods for Economic Theory*.
- **J. H. Boyd III**: *Mathematics for Economists* and Microeconomics I materials.
- **K. C. Border**: online notes repository.
- **F. Echenique**: *Convexity, Constrained Maximization, Envelope Theorem, etc.*
- **T. Sarver**: teaching materials, Duke.
- **P. Schrimpf**: *Mathematics for Economics*, UBC 526.
- **S. Boyd & L. Vandenberghe**: *Convex Optimization* slides.
- **A. Reshef**: *Introduction to Mathematical Economics I*.

♦ Closest in content to this course.

Resources

SELECTED VIDEO RESOURCES

Trefor Bazett: calculus, multivariable calculus, and mathematical foundations.

3Blue1Brown: visual intuition for linear algebra, calculus, and probability.

TrevTutor: logic, proof methods, and functions.

Harvard Stat 110 (Joe Blitzstein): probability.

♦ **Arizona Math Camp**: mathematical methods for economists.

Stanford EE364A (Stephen Boyd): convexity, duality, and optimization.

Ben Lambert: econometrics, OLS, and statistical inference.

♦ Closest in content to this course.